

RUSH CITY, MN, USA

WMO: 726679

Lat: 45.698N

Lon: 92.953W

Elev: 281

StdP: 97.99

Time Zone: -6.00 (NAC)

Period: 01-19

WBAN: 04969

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-26.0	-22.9	-29.9	0.2	-25.3	-27.5	0.3	-22.5	8.4	-7.6	7.6	-7.6	1.5	260	0.537

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.7	31.2	22.6	29.2	21.5	27.7	20.5	24.0	28.9	22.9	27.7	21.8	26.3	4.0	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
22.4	17.7	27.3	21.3	16.5	26.2	20.2	15.4	24.9	73.4	29.2	69.2	27.7	65.1	26.2	27.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.2	7.3	6.4	-29.7	34.0	2.9	2.0	-31.8	35.4	-33.5	36.6	-35.2	37.7	-37.3	39.2		
(5)			-29.9	25.7	2.9	1.1	-32.0	26.5	-33.7	27.2	-35.3	27.8	-37.3	28.6	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.4	-9.9	-8.4	-1.1	6.5	12.9	18.2	21.1	19.5	15.5	7.9	0.6	-6.6	(6)	
	DBStd	12.05	7.32	6.66	6.98	5.48	4.74	3.43	3.02	3.21	4.79	5.07	5.71	6.28	(7)	
	HDD10.0	2543	616	514	348	132	23	0	0	0	8	102	284	516	(8)	
	HDD18.3	4576	874	748	602	357	180	42	9	23	111	326	531	774	(9)	
	CDD10.0	1243	0	0	5	26	112	247	343	296	174	37	3	0	(10)	
	CDD18.3	233	0	0	0	1	11	39	93	61	26	2	0	0	(11)	
	CDH23.3	2242	0	0	2	16	143	392	890	543	233	23	0	0	(12)	
	CDH26.7	596	0	0	0	3	39	90	257	143	61	2	0	0	(13)	
(14) Wind	WSAvg	2.6	2.7	2.8	2.9	3.3	3.0	2.4	2.1	1.9	2.4	2.7	2.8	2.6	(14)	
(15) Precipitation	PrecAvg	805	22	22	39	71	90	110	103	103	93	79	45	28	(15)	
	PrecMax	1004	55	47	74	211	180	183	183	277	172	202	120	54	(16)	
	PrecMin	591	2	2	15	16	25	62	37	35	23	28	1	8	(17)	
	PrecStd	125	15	13	17	40	43	32	38	53	47	45	35	15	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.6	9.9	20.6	25.9	31.0	32.1	33.3	32.7	31.3	26.4	18.0	7.7	(19)	
		MCWB	3.3	6.7	14.0	16.4	20.1	23.4	24.5	23.3	22.0	17.4	12.4	5.3	(20)	
	2%	DB	3.1	5.1	15.4	21.8	27.1	29.0	31.2	30.1	28.0	22.6	14.7	4.9	(21)	
		MCWB	1.1	2.3	10.8	12.7	17.7	20.7	23.4	22.3	20.8	15.4	10.1	3.1	(22)	
	5%	DB	1.4	2.7	11.7	18.8	24.2	27.3	29.3	27.9	26.2	18.9	11.7	2.7	(23)	
		MCWB	0.3	0.6	7.9	10.8	15.9	19.9	22.1	20.8	19.5	13.4	7.8	1.3	(24)	
	10%	DB	0.1	0.9	7.9	16.3	21.9	25.7	27.8	26.4	23.8	16.2	8.9	1.3	(25)	
		MCWB	-0.9	-0.6	4.8	9.2	14.9	18.6	21.0	19.9	18.2	11.8	6.2	0.4	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.5	6.7	15.6	16.9	22.3	24.6	26.2	24.6	23.9	19.7	13.5	6.6	(27)	
		MCDB	6.1	10.5	19.0	24.7	28.2	29.7	31.2	30.4	28.2	22.8	16.6	7.5	(28)	
	2%	WB	1.6	2.8	11.3	13.8	19.1	22.6	24.4	23.4	21.9	16.3	10.9	2.9	(29)	
		MCDB	3.0	4.8	14.6	19.5	24.4	27.5	29.3	28.2	26.6	20.9	14.2	4.4	(30)	
	5%	WB	0.4	0.8	7.9	11.7	17.6	20.9	23.3	22.1	20.5	14.2	8.0	1.6	(31)	
		MCDB	1.4	2.4	11.6	17.6	22.3	25.4	28.0	26.4	24.7	18.7	10.6	2.7	(32)	
	10%	WB	-1.0	-0.5	5.2	10.0	15.8	19.6	22.1	20.9	19.1	12.1	6.2	0.4	(33)	
		MCDB	-0.1	0.6	7.9	15.6	20.4	24.2	26.6	25.1	23.2	15.7	8.9	1.2	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	8.9	10.0	10.1	11.4	12.2	11.6	11.7	11.7	11.8	10.1	8.6	7.6	(35)	
		MCDBR	9.0	10.5	13.3	16.4	15.8	13.4	13.1	13.5	13.8	14.6	12.9	8.4	(36)	
	5% WB	MCWBR	7.5	8.2	8.6	9.4	8.0	6.9	6.6	6.7	7.2	8.4	8.6	6.9	(37)	
		MCWBR	8.5	9.8	12.6	14.3	13.1	11.6	11.8	11.9	11.9	12.6	11.7	7.5	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.258	0.269	0.316	0.371	0.404	0.389	0.384	0.396	0.368	0.316	0.287	0.263	(40)	
		taud	2.398	2.361	2.358	2.304	2.258	2.335	2.375	2.352	2.405	2.539	2.513	2.421	(41)	
	Ebn at Noon		865	920	916	887	865	878	878	853	848	857	825	817	(42)	
		Edh at Noon	70	89	105	122	132	123	117	115	100	76	64	62	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.68	2.77	3.88	4.64	5.33	5.89	6.23	5.20	4.00	2.49	1.57	1.17	(44)	
	RadStd		0.18	0.26	0.34	0.44	0.43	0.43	0.21	0.39	0.28	0.33	0.14	0.14	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (60 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page